Chapter 1

Introduction

In recent years, the term *quantum materials* has come to describe a class of materials that, loosely speaking, display quantum mechanical properties and many-body effects over large length and energy scales. These effects manifest in the form of exotic phenomena, examples of which include unconventional superconductivity and the fractional quantum hall effect. Such materials often involve strong inter-electron interactions, low dimensions and topological properties, making them highly sensitive to small changes in parameters or temperature. This typically means that such materials cannot be described using single-particle or weak-coupling approaches, making the study of such materials more challenging and interesting. Examples of quantum materials include the copper oxide superconductors, heavy fermion materials, Moire materials and quantum spin liquids. The exotic phases that emerge from the interplay of competing tendencies in these materials are often referred to as *quantum matter*. Such phases are usually marked by non-trivial patterns of topology and entanglement. There are several examples of phases of quantum matter, some were already provided: unconventional superconducting states, fractional quantum hall states, strange metals and topologically ordered phases, among others. In this thesis, we are concerned with the study of a specific aspect of quantum matter – Mott metal-insulator transition. In doing so, we will deal with a number of correlated phases such as Mott insulators, strange metals and the pseudogap, all of which will be explained in due course. The phenomenon of Mott insulation is also intimately tied to other exotic effects such as high-T_c superconductivity and spin liquid behaviour; these will not be explored in this thesis.

1.1 Mott metal-insulator transition: An Introduction

The rich physics of Mott metal-insulator transitions in strongly correlated systems has been an active subject of study for quite some time [1–4]. Their appearance in various materials such as the nickelates, cuprates, iridates and manganites, among others, is marked by complex phase diagrams that display dramatic changes in properties across changes in filling and/or bandwidth, even at low temperatures. The physics of Mott transitions is driven by the competition between two energy scales in the system: (i) the scale of itineracy, quantified by the bandwidth, and (ii) the scale of electron-electron repulsion. A Mott insulator arises when the electronic correlations are strong enough that low-energy electronic excitations become gapped, and the metallic valence band gets split into lower and upper bands; for an initially half-filled valence band, the lower band is filled at $T = 0$ while the upper band is vacant. Because of strong electronic repulsion, the electrons in the lower band form local moments which can show a variety of behaviour. This includes various symmetry-broken ground states such as an antiferromagnet [5], or symmetry-preserved states such as quantum spin liquids [6]. The physics of Mottness has been proposed as being responsible for several very interesting phenomena, such as high-temperature superconductivity [5], non-Fermi liquids [7] and the various phases in magic-angle twisted bilayer graphene [8, 9].

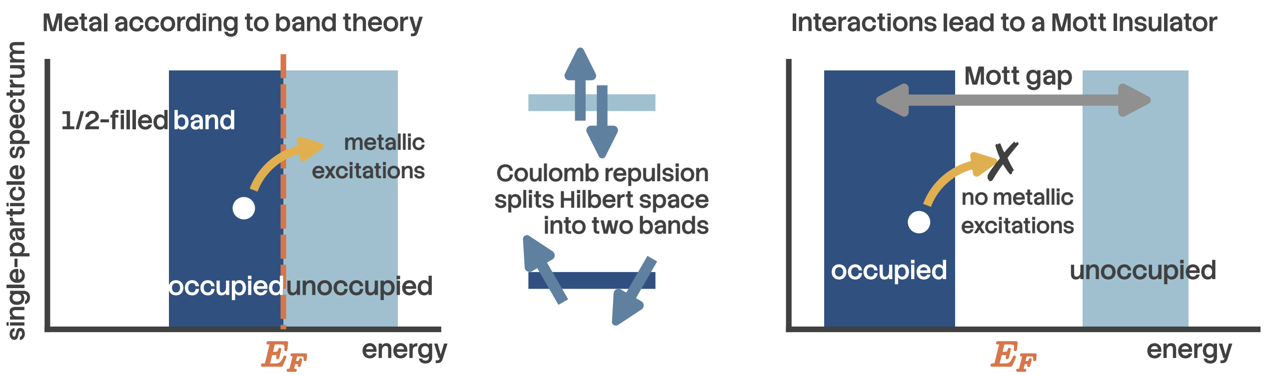

Figure 1.1

Initial attempts at explaining metal-insulator transitions were based on a non-interacting picture of electrons (through the work of Bethe, Sommerfeld, Bloch and Wilson among others); in such a picture, electronic bands can arise only from the scattering of electrons off the periodic lattice potential. One then distinguishes between metals and insulators at zero temperature by the position of the Fermi level E_F with respect to the bands: if E_F lies within a band gap, an integer number of bands will be completely filled and the system will be an insulator, whereas if E_F lies within one of the bands, the system will be a metal because there are vacant gapless excitations within the band.

It was first pointed out by Boer and Verwey [10] that certain insulators and semiconductors (such as nickel oxides) have a particle filling that, purely from band theory, would lead to partially filled bands. This poses a challenge because a partially filled band necessarily leads to a metallic phase (see Fig. 1.1 (left)). The insulating behaviour of these materials therefore signals a fundamental limitation of single-particle band theory and points to a qualitatively different mechanism for electron localization—one driven by strong electron–electron interactions rather than by scattering from the lattice potential alone. This realization led to the concept of a Mott insulator, in which charge localization emerges as a genuine many-body effect despite the absence of a band gap in the non-interacting electronic structure. A very simple mechanism for this was put forth by Mott [11–13]: in the presence of a large local Coulomb repulsion cost U on a lattice site, the quantum states $|\Psi_1\rangle$ in which the site is singly occupied get separated in energy from the states $|\Psi_2\rangle$ where the site is doubly occupied by a gap of size U; the former class of states $|\Psi_1\rangle$ form the lower band (LB) while the latter $|\Psi_2\rangle$ form the upper band (UB). If we now have a half-filled system (one electron per site), this electron will completely fill LB and leave UB vacant, leading to an insulating state (see Fig. 1.1 (right)).

Given the above discussion, it is clear that a Mott insulator must involve a gap to charge excitations (excitations that add or remove an electron). It does not however determine the fate of the spin, orbital or other degrees of freedom. One scenario is when the spin or orbital degree of freedom undergoes spontaneous symmetry breaking into an ordered state; in the case of spin, this usually results in a superexchange interaction being generated and leads to an long-range ordered antiferromagnetic Neel state coexisting with the Mott state. Examples of such systems include vanadium sesquioxide V_2O_3 and its derivatives and the high-T_c cuprates (such as La_2CuO_4) [4]. The other scenario is one where the spin degree of freedom retains its symmetry, and we end up with a quantum mechanically non-trivial entangled state (in order to quench the entropy). Examples of such states include quantum spin liquids [14]. Mott insulators without long-range order are of particular interest because of the possibility of topological order, fractional excitations, and novel metallic parent states.

A prototypical model for gaining theoretical understanding of Mott metal-insulator transitions on a lattice is the single-band Hubbard model [1, 15–17]

$$H = -t \sum_{\langle i,j \rangle,\sigma} (c^\dagger_{i,\sigma} c_{j,\sigma} + \text{h.c.}) - \frac{U}{2} \sum_i (n_{i,\uparrow} - n_{i,\downarrow})^2 - \mu N \ . \tag{1.1}$$

The first term incorporates nearest-neighbour hopping on the lattice, while the second term introduces inter-electron repulsion when two electrons site on the same lattice site; in this way, the model incorporates the two competing tendencies of delocalisation and localisation that are responsible for the Mott MIT. The third term fixes the chemical potential and determines the filling of the lattice. Many of the materials that are studied for Mott MIT involve $d-$electrons that are five-fold degenerate, and these orbitals can hybridise with ligand atoms. The Hubbard model avoids these complexities by focusing on only a single band that lies closest to the Fermi energy, and it is the states of this band that are described by the Hubbard model.

This leaves us with two channels for observing Mott MIT [4]: (i) tuning the bandwidth, and (ii) tuning the filling. The former can be done, in experiments, by applying pressure. Filling is tuned by doping specific elements of a compound with elements of a different valence. The former is referred to as bandwidth-control metal-insulator transition, while the latter is called filling-control metal-insulator transition.

1.2 Features of Mott metal-insulator transition

1.2.1 Two complementary pictures of the Mott transition: Brinkman-Rice vs Mott-Hubbard.

Initial insights into the Mott MIT were obtained by Hubbard [1], thinking along lines similar to Mott. Within this picture, one starts from the insulating large U limit of the Hubbard model (eq. 1.1) ($U \gg t$. We stay at half-filling for simplicity ($\mu = 0$). The large on-site repulsion leads to the formation of two bands at each site, the lower Hubbard band (LHB) associated with singly-occupied states and the upper Hubbard band (UHB) associated with doubles and holes (states that incur a cost U). In the large U limit, these bands are well-separated, and the dynamics within the bands is a result

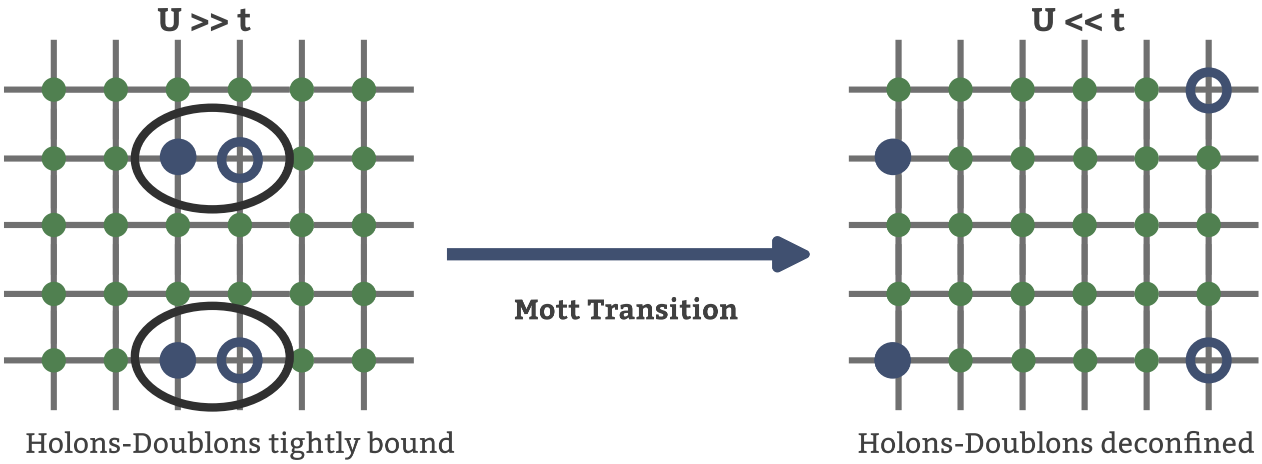

Figure 1.2

of the motion of tightly bound holons and doubles (more on this later) that are generated because of the small but non-zero hopping probablity t. As U/t is lowered, the bands come closer and a metal-insulator transition occurs at the critical value of U at which the gap vanishes. For lower U/t, the bands overlap and we get a metal. This picture fails to provide an accurate description of the metal resulting from the melting of such a Mott insulator [18].

An alternative approach to the Mott MIT is due to Brinkman and Rice [19], where the authors applied Gutzwiller's variational method [20] to the half-filled Hubbard model. Gutzwiller's approach involves modifying, through variational factors, the hopping matrix elements that start from doubly occupied states, in order to reflect the electronic correlation at each site. By requiring that the number of doubly-occupied states vanish in the Mott insulating ground state, they obtained a critical value of the interaction strength. In the Brinkman-Rice scenario, therefore, the transition from the metal to the insulating phase occurs when the hopping matrix elements from the doubly-occupied to the singly-occupied states strictly vanishes. Note that in contrast to the Mott-Hubbard picture, this approach starts from a correlated Fermi liquid (renormalised due to the Gutzwiller factors). While it gives a better low-energy description of the correlated metal compared to the Mott-Hubbard picture, it fails to account for the high-energy physics, and treats the Mott insulator as a direct product state of local moments at each state (on account of the strictly zero hopping). This of course means that this method cannot explain antiferromagnetic superexchange processes in the Mott insulator [18].

1.2.2 Holon-doublon unbinding picture of the Mott MIT.

In 1949, Mott proposed a zero temperature explanation for a correlation-driven insulator and a mechanism for its transition into a metal in terms of the motion of holes and doubles in the background of a half-filled lattice (see Fig. 1.2). For $U \ll t$ (strong correlation), the ground state predominantly involves lattice sites with a single electron. Hopping processes between two nearest-neighbour sites transfer an electron between them and create a pair of hole and double, but the pair remains tightly bound to each other. This is because moving the hole independently involves scattering processes that cross the Mott gap, and these processes are not allowed at $T = 0$. But without the independent motion of the hole (or double), there is no current, and the system is an insulator (left, Fig. 1.2).

As U/t is lowered, the gap closes and processes that transport the hole without transporting the double become gapless. This essentially means that the pair becomes deconfined, and the hole can be moved far away from the double (right, Fig. 1.2). Current can then be passed by having the hole move through the lattice, and we get a metallic state. This leads to a picture of the Mott transition into a metal in terms of an unbinding of the hole-double bound states (excitons) that were present in the insulator.

There have been more quantitative studies of this mechanism. I list some of the relatively recent ones here. In [21], Prelovšek et al. studied the motion of holon-doublon pairs in a spin polarised background by rewriting the repulsive Hubbard model in terms of electrons as an attractive Hubbard model of holes and doubles. The model was treated using a BCS-like treatment, and the Mott transition was obtained when the binding energy for the hole-double condensate vanishes. To mark the deconfinement, they show that the approach to the transition from the insulating side is concomitant with an increase in the size of the pairs. This approach however fails to capture the transition for a system away from half-filling. Following a slave-boson treatment of holon-doublon unbinding, Zhou et al. [22] show that the Mott transition from a paramagnetic metal to a Mott insulator on the half-field honeycomb and 2D square lattices occur through an intervening Slater insulator with coherent quasiparticles. In particular, their approach sheds light on the nature of the incoherent excitations in the Hubbard sidebands of the local spectral function. Finally, an older work by Watanabe et al. [23] studies the half-filled Hubbard model on an anisotropic triangular lattice using a Monte carlo method with a variational binding parameter between holons and doublons. Their approach reveals a first order Mott transition, along with robust $d-$wave superconductivity and short-range magnetic order.

1.2.3 Spectral features of the Mott transition.

One of the important diagnostics of the Mott metal-insulator transition is the spectrum for one-particle electron-like excitations, as expressed through the one-particle retarded Greens function $G(\boldsymbol{k},\sigma;t) = -i\theta(t)\langle\{c_{\boldsymbol{k},\sigma}(t), c^\dagger_{\boldsymbol{k},\sigma}\}\rangle$. In general, the presence of poles in the Green function at the Fermi energy and in its neighbourhood imply the existence of gapless excitations and hence a metallic phase. It turns out that this statement is equivalent to Luttinger's theorem [24, 25],

$$\frac{N}{V} = 2\int_{G(0,\boldsymbol{k})>0} \frac{d\boldsymbol{k}}{(2\pi)^3} , \tag{2.1}$$

which states that the particle density in a metal is equal to the number of states in the Fermi volume. As clarified in the equation, the Fermi volume is defined as the region of $k-$space where the Greens function is positive at $\omega = 0$.

In a Mott insulator, strong correlations lead to the destruction of electronic excitations around $\omega = 0$. This results in the Greens function poles transmuting into zeros. This is also tracked by analogous changes in the electronic self-energy $\Sigma(\boldsymbol{k};\omega)$; Greens function zeros leads to a Mott pole at $\omega = 0$ in $\Sigma(\boldsymbol{k};\omega)$. The points in $k-$space where $\Sigma(\boldsymbol{k};0)$ vanishes is called a Luttinger surface [26]. The Mott metal-insulator transition is therefore about the topological change of a

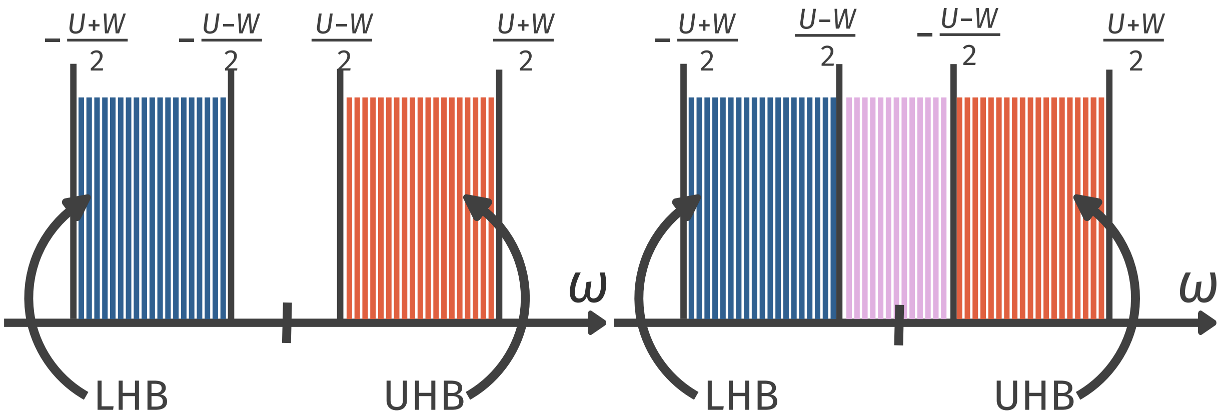

Figure 1.3

Fermi surface (surface of Greens function poles) into a Luttinger surface (surface of self-energy poles).

As discussed earlier, the appearance of Greens function zeros at $\omega = 0$ leads to the formation of a charge gap that freezes one-particle low-energy excitations. The spectrum gets redistributed into two bands on either side of the gap, the upper and lower Hubbard bands. In the ground state, the lower band is obviously filled while the upper band is empty. Bands of a Mott insulator formed from electronic correlation differ from those formed due to one-particle scattering in a particular way: dynamical spectral weight transfer. Unlike a band insulator, the bands of a Mott insulator are not rigid; tuning the lattice filling results in a continuous transfer of states from one band to another.

The non-rigidity of the bands can be seen most easily from the atomic limit ($t = 0$) of the Hubbard model (eq. 1.1). The local retarded Greens function at an arbitrary site i can be obtained using the equation of motion approach. The final result is [27]

$$G_R(i;\omega) = \frac{1+x}{\omega+\mu+U/2} + \frac{1-x}{\omega+\mu-U/2}\ , \tag{2.2}$$

where $x = 1 - \sum_\sigma \langle n_{i\sigma}\rangle$ is the average number of holes at any site. The spectral weight for the two poles makes it clear that the lower band has $1 + x$ number of states while the upper band has $1 - x$ states. Changing the number of states in the lower band (by tuning μ and hence x) also changes those in the upper band. For example, by adding 0.1 holes per site to the lower band, the spectral weight in the upper band decreases by the same amount. This is the non-rigidity of the bands mentioned previously.

1.2.4 Preliminary Insights into Mottness From a Toy Model

The Baskaran-Hatsugai-Kohmoto model [28, 29] introduced independently in 1991 and 1992 is an exactly solvable model of electronic correlation that displays a transition from a metallic phase to a Mott insulator. While it's solvability offers an instructive method for studying Mott transition, the infinitely long-ranged nature of its interaction makes it tricky to apply these conclusions to realistic materials.

At half-filling, the model is defined by the following Hamiltonian:

$$H_{\text{BHK}} = \sum_{\mathbf{k},\sigma} \epsilon(\mathbf{k}) n_{\mathbf{k},\sigma} - \frac{U}{2} \sum_{\mathbf{k}} (n_{\mathbf{k},\uparrow} - n_{\mathbf{k},\downarrow})^2 \ . \tag{2.3}$$

Clearly the model is diagonal in momentum-space; the Hamiltonian splits up into commuting 4×4 blocks for every point $\mathbf{k}$:

$$H_{\text{BHK}} = \sum_{\mathbf{k}} H(\mathbf{k}), \quad H(\mathbf{k}) = \epsilon(\mathbf{k}) \sum_{\sigma} n_{\mathbf{k},\sigma} - \frac{U}{2} (n_{\mathbf{k},\uparrow} - n_{\mathbf{k},\downarrow})^2 \ . \tag{2.4}$$

The vacant eigenstate $|0\rangle$ of $H(\mathbf{k})$ has zero energy ($E_0 = 0$), the two singly-occupied states $|\uparrow\rangle$ and $|\downarrow\rangle$ have energy $E_1 = \epsilon(\mathbf{k}) - U/2$ and the doubly-occupied state $|\uparrow, \downarrow\rangle$ has energy $E_2 = 2\epsilon(\mathbf{k})$. This leads to two possible single-particle excitations – one for adding an electron on the vacant state: $(\Delta_1 = E_1 - E_0 = \epsilon(\mathbf{k}) - U/2$, the other for adding an additional electron on a singly-occupied state: $\Delta_2 = E_2 - E_1 = \epsilon(\mathbf{k}) + U/2$.

Upon combining states from other $k-$modes, these isolated excitations broaden into bands around $\pm U/2$ because of the continuum of values available to $\epsilon(\mathbf{k})$. For large enough U (compared to the non-interacting bandwidth W), these bands remain well-separated in energy (Fig. 1.3 left), forming lower and upper Hubbard bands. The half-filled system results in the lower band being completely filled and the upper band empty, leading to a Mott insulator. The states within this band are all singly-occupied (per $k-$state). Upon lowering the correlation strength U, the bands touch each other at the critical value of $U_c = W$ (obtained by equating the top of the lower band to the bottom of the upper band). This mechanism is qualitatively similar to the Mott-Hubbard picture of Mott transition. For lower values of U, the system is a metal due to the presence of gapless excitations within the combined band (Fig. 1.3 right). This therefore provides a simple demonstration of a correlation-driven Mott metal-insulator transition.

1.3 Models and materials for strong local correlation

1.3.1 Introduction to the auxiliary model approach

One of the main results obtained within this thesis is the development of a formalism to study the more realistic case of finite-dimensional interacting models by using appropriately chosen impurity models as auxiliary models. The key to an *auxiliary model method* is to construct an appropriate auxiliary model that while being tractable captures some essential property of the full lattice model [30]. This involves separating the complete degrees of freedom into a system part (H_S) which one treats exactly and the rest of the system (H_R) that one might simplify in order to be able to solve the problem. The non-trivial nature of the problem arises of course from the hybridisation H_{SR} between the system and the bath:

$$H = \begin{bmatrix} H_R & H_{RS} \\ H_{RS}^* & H_S \end{bmatrix} . \tag{3.1}$$

This separation can always be done exactly, if one does not care about tractability. For example, one can take the Hubbard model that describes electrons hopping on a lattice and interacting when two electrons reside on the same lattice site:

$$H_{\text{hub}} = -t \sum_{\langle i,j\rangle,\sigma} \left(c^\dagger_{i,\sigma} c_{j,\sigma} + \text{h.c.}\right) + U \sum_i n_{i\uparrow} n_{i\downarrow} \ , \tag{3.2}$$

and separate this into parts, the system being the local orbitals of a specific site l and the rest of the system being all the other sites:

$$H_S = U n_{l\uparrow} n_{l\downarrow}, \quad H_R = -t \sideset{}{'}\sum_{\langle i,j\rangle,\sigma} \left(c^\dagger_{i,\sigma} c_{j,\sigma} + \text{h.c.}\right) + U \sideset{}{'}\sum_{i} n_{i\uparrow} n_{i\downarrow}, \quad H_{SR} + H_{RS} = -t \sum_{i,\sigma} \left(c^\dagger_{l,\sigma} c_{i,\sigma} + \text{h.c.}\right) \ , \tag{3.3}$$

where the primed summation is carried out over all nearest-neighbour pairs that do not involve l and the summation in $H_{SR} + H_{RS}$ is over all sites adjacent to l.

The smaller system H_S (often called the *cluster*) is typically chosen such that its eigenstates are known exactly. Progress is then made by choosing a simpler form for H_R and its coupling H_{RS} with the smaller system. This combination of the cluster and the simpler bath is then called the *auxiliary system*. A tractable auxiliary system for the Hubbard model is the single-impurity Anderson model (SIAM); the correlated impurity site represents the set of local orbitals of any given site, while the conduction bath represents the rest of the lattice sites in an approximate fashion. Such a construction is shown in fig. 1.4.

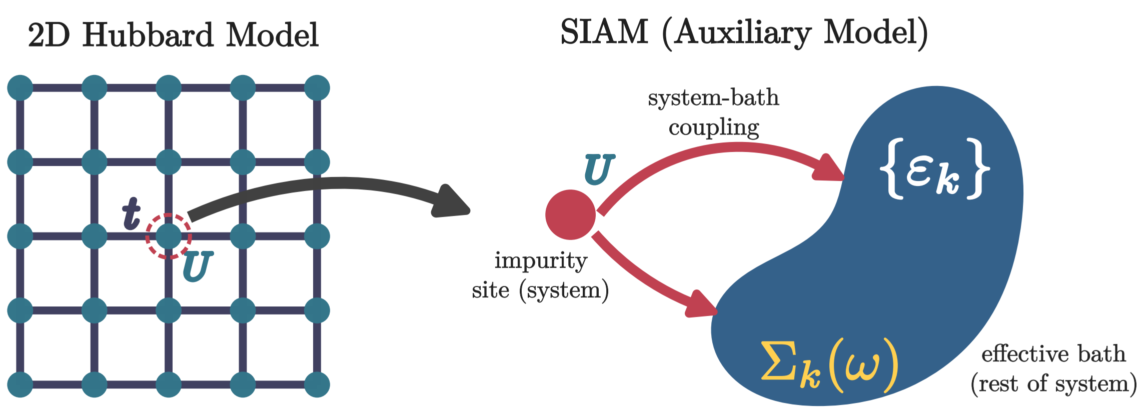

Figure 1.4: *Left: Full Hubbard model lattice with onsite repulsion U^H on all sites and hopping between nearest neighbour sites with strength t^H. Right: Extraction of the auxiliary (cluster+bath) system from the full lattice. The central site on left becomes the impurity site (red) on the right (with an onsite repulsion ϵ_d), while the rest of the $N-1$ sites on the left form a conduction bath (green circles) (with dispersion ϵ_k and correlation modelled by the self-energy $\Sigma_k(\omega)$) that hybridizes with the impurity through the coupling V.*

It should be noted that any reasonable choice of the cluster and bath would break the translational symmetry of the full model: To allow computing quantities, one would need to make the bath (which is a much larger system) simpler than the cluster (which is a single site). This distinction breaks the translational symmetry of the Hubbard model. As a result, calculations that make use of the translation symmetry of the full model (such as $k-$space objects) will require a periodisation operation of an auxiliary model quantity.

1.3.2 Correspondence between impurity and lattice models in $d = \infty$

Dynamical Mean-Field Theory (DMFT) provides a nonperturbative framework for treating strong local electronic correlations in lattice models by mapping them onto a self-consistent quantum impurity problem [18, 31]. The central idea of DMFT is that in the limit of infinite spatial dimension, or equivalently infinite coordination number, the electronic self-energy becomes purely local while retaining its full dynamical (frequency-dependent) structure. As a result, the many-body lattice problem reduces to an effective single-site problem embedded in a self-consistently determined bath, capturing local quantum fluctuations exactly while neglecting nonlocal correlations. DMFT has become a cornerstone of modern correlated-electron theory, successfully describing Mott metal–insulator transitions, quasiparticle formation, and spectral weight transfer in Hubbard-like models [18, 32].

In order to motivate the impurity model-lattice model mapping inherited by DMFT, we first recall the textbook example of classical mean-field theory. Mean-field theory provides one of the earliest and most instructive examples of controlled approximations in many-body physics. In its classical incarnation, exemplified by the Ising model,

$$H_{\text{Ising}} = -\sum_{i\neq j} J_{ij} S_i S_j - h\sum_i S_i = -\sum_i S_i (h + \sum_{j\neq i} J_{ij} S_j), \tag{3.4}$$

mean-field theory proceeds by replacing the interaction between a given spin and its neighbors by an *effective static field* proportional to the average magnetization. Writing $S_j = \langle S_j\rangle + \delta S_j$ and neglecting quadratic fluctuations and constant parts, one obtains an *auxiliary problem* in the form of a mean-field Hamiltonian

$$H^{\text{(MF)}}_{\text{Ising}} = -\sum_i (h + h_{\text{MF}}(i)) S_i \ , \tag{3.5}$$

where the mean-field h_{MF} comes out to be

$$h_{\text{MF}}(i) = \sum_{j\neq i} 2\langle S_j\rangle J_{ij} \ . \tag{3.6}$$

The above approach assumes two things: (i) the presence of a non-zero *order parameter* $m_j = \langle S_j\rangle$ (in this case, a non-zero magnetisation), and (ii) the smallness of the fluctuations $S_j - \langle S_j\rangle$. The mean-field Hamiltonian can be used to compute the partition function (at inverse temperature β) and hence the magnetisation at a site i of the lattice:

$$m_i = \tanh(\beta(h_{\text{MF}}(i) + h)) = \tanh(\beta \sum_{j\neq i} 2 m_j J_{ij} + \beta h) \ . \tag{3.7}$$

The problem hasn't yet been solved of course; one needs to determine the appropriate value of m_j. We therefore need an additional equation; this is provided by the so-called *self-consistency* *condition*. The translation invariance of the lattice ensures that the magnetisation must be uniform across sites i and j. This allows us to replace m_j with m_i in the above equation:

$$m_i = \tanh(\beta m_i \sum_{j \neq i} 2J_{ij} + \beta h) \ . \tag{3.8}$$

This equation can be solved numerically to obtain the uniform magnetisation $m_i = m$. The mean-field approximation becomes exact in the limit of large coordination number, where fluctuations are suppressed and spatial correlations beyond a single site vanish.

An equivalent way of framing the self-consistency equation is to equate the mean-fields on different lattice sites. Combining eqs. 3.7 and 3.6, we have

$$h_{\mathrm{MF}}(i) = \sum_{j \neq i} 2m_j J_{ij} = \sum_{j \neq i} 2\tanh(\beta(h_{\mathrm{MF}}(j) + h))J_{ij} \ . \tag{3.9}$$

The equation in terms of $h_{\mathrm{MF}}(i)$ and $h_{\mathrm{MF}}(j)$ can again be solved numerically by claiming that the mean-field has to be uniform across all the sites owing to translation invariance. This form of the self-consistency requirement (in terms of the field) will be useful to us in the upcoming discussion.

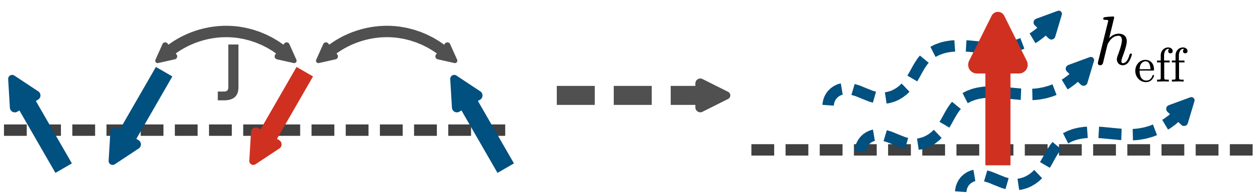

Figure 1.5

The essential feature of classical mean-field theory is therefore the replacement of a many-body problem by an effective *single-site* problem embedded in a self-consistently determined environment. Quantum lattice models admit a closely analogous construction, but with an important conceptual difference: while the classical mean field is static, quantum fluctuations necessarily introduce a tower of excited states that allow for temporal dynamics. This observation underlies the formulation of dynamical mean-field theory (DMFT), which may be viewed as the natural quantum generalization of classical mean-field theory [18, 31].

Similar to the previous case of the Ising model, we will now derive the self-consistency equation for a quantum model. Consider an interacting fermionic lattice Hamiltonian, such as the Hubbard model:

$$H = -\sum_{ij\sigma} t_{ij} c^\dagger_{i\sigma} c_{j\sigma} + U \sum_i n_{i\uparrow} n_{i\downarrow}, \tag{3.10}$$

defined on a lattice with coordination number z.

The point of this exercise is to calculate the interacting self-energy $\Sigma_k(\omega)$ for the fermions. This therefore acts as an "order parameter" and plays the same role as the local magnetisation in the previous exercise. The self-energy defines the interacting k-space Green's function for a translation-invariant problem:

$$G_{\mathbf{k}}(\omega) = \frac{1}{\omega - \varepsilon_{\mathbf{k}} - \Sigma_{\mathbf{k}}(\omega)}, \tag{3.11}$$

where $\varepsilon_{\mathbf{k}}$ denotes the noninteracting dispersion, and a real-space local Green's function, obtained by summing over momentum:

$$G_{\text{loc}}(\omega) = \sum_{\mathbf{k}} \frac{1}{\omega - \varepsilon_{\mathbf{k}} - \Sigma_{\mathbf{k}}(\omega)}. \tag{3.12}$$

Similar to the earlier exercise, we need to apply a mean-field decoupling to drop certain fluctuations and obtain a similar effective problem. In the context of interacting electrons, such a decoupling is obtained in the limit of infinite dimensionality or coordination number, where the self-energy becomes $k-$independent [31]:

$$\Sigma_{\mathbf{k}}(\omega) = \Sigma_{\text{loc}}(\omega), \text{ or (equivalently) } \Sigma_{ij}(\omega) = \delta_{ij}\Sigma_{\text{loc}}(\omega)\ . \tag{3.13}$$

The mean-field decoupling here therefore corresponds to ignoring spatial fluctuations of the one-particle self-energy, and leads to a local Greens function that is directly connected to the local self-energy:

$$G_{\text{loc}}(\omega) = \sum_{\mathbf{k}} \frac{1}{\omega - \varepsilon_{\mathbf{k}} - \Sigma_{\text{loc}}(\omega)}. \tag{3.14}$$

It turns out that a local Greens function connected to a local self-energy also defines a quantum impurity problem describing the dynamics of a single correlated site with the other environment sites integrated out:

$$S_{\text{imp}} = -\int_0^\beta d\tau d\tau'\, c^\dagger(\tau)\, G_{\text{MF}}^{-1}(\tau - \tau')\, c(\tau') + U \int_0^\beta d\tau\, n_\uparrow(\tau) n_\downarrow(\tau). \tag{3.15}$$

This acts as an auxiliary problem for the more complete problem of locally interacting electrons on a lattice. The non-interacting Greens function G_{MF} is as of now unknown, and defines the environment of the impurity site and comprises the dynamics of the sites that have been integrated out. Such an impurity problem is solvable, and the impurity Green's function is given by

$$G_{\text{imp}}(\omega) = \frac{1}{G_{\text{MF}}^{-1}(\omega) - \Sigma_{\text{imp}}(\omega)}, \tag{3.16}$$

The first term in eq. 3.15 is therefore what makes this problem dynamical – it represents the probability of electrons hopping into and out of the impurity site, and leads to fluctuations of its occupancy. We can make the identification of the Greens function G_{MF} as the *mean-field* encountered in the previous example. More specifically, G_{MF} acts as a “field” because it is obtained by integrating out the other sites on the lattice, and it acts on the impurity site by affecting its dynamics.

The complete set of equations for the solving such a dynamical mean-field problem is finally obtained by equating the impurity local Greens function G_{imp} to the lattice local Greens function G_{loc}. As discussed earlier, such a solution exists in an exact fashion only in the limit of infinite dimensions, as long as the correct G_{MF} can be found. In order to find the appropriate G_{MF} that allows equating the impurity model Greens function to a lattice mode Greens function, we require an additional equation: the self-consistency condition. Similar to the case of the Ising model, we require that G_{MF} be equal to G_{loc}, owing to translation invariance. The full set of coupled DMFT equations is

$$\begin{aligned} G_{\text{MF}} &= G_{\text{loc}}, \\ \Sigma_{\text{imp}}(\omega) &= G_{\text{MF}}^{-1}(\omega) - G_{\text{imp}}^{-1}(\omega), \\ \Sigma_{\text{loc}}(\omega) &= \Sigma_{\text{imp}}(\omega), \\ G_{\text{loc}}(\omega) &= \sum_{\mathbf{k}} \frac{1}{\omega - \varepsilon_{\mathbf{k}} - \Sigma_{\text{loc}}(\omega)}. \end{aligned} \tag{3.17}$$

One can map various features of this self-consistent program with the set of equations obtained above for the Ising model:

- The first equation is equivalent to the feature in the previous exercise where we equated the mean-fields on i and j.
- The second equation is equivalent to computing, in the previous exercise, the magnetisation on the j^{th} site from the partition function.
- The third equation is the essential mean-field approximation, where we equate the self-energy of the simpler decoupled problem to that of the more complete lattice problem and is justified in the limit of infinite dimensions.
- The fourth equation is equivalent, in the previous exercise, to using the computed "magnetisation" of site j in the second step to further compute the mean-field at site i.

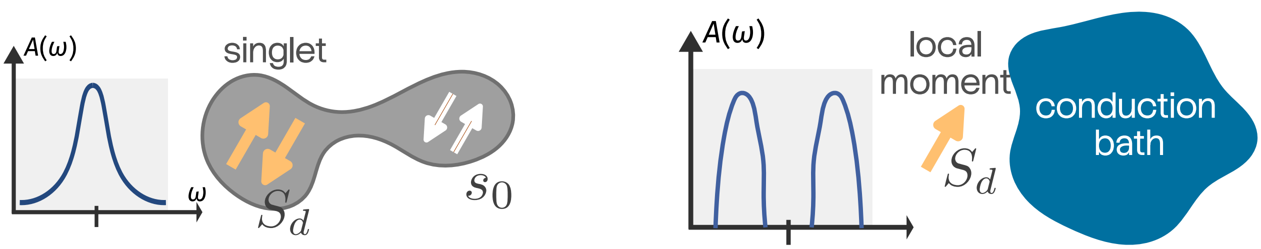

Figure 1.6

In direct analogy with classical mean-field theory, DMFT replaces a lattice model by an effective single-site problem embedded in a self-consistent environment. The crucial distinction is that the mean field is now dynamical rather than static, encoding quantum fluctuations exactly while neglecting spatial correlations. The fact that the correlated self-energy of the impurity site feeds into the impurity-bath hybridisation during the convergence towards self-consistently means that the bath must get correlated in the process. Importantly, this demonstrates the importance of quantum impurity models in investigating correlated lattice models, particularly in the presence of strong local interactions. This is because the effects of local correlations are captured exactly by such an auxiliary model mapping.

1.3.3 Examples of materials with strong local correlations

As discussed in the previous section, the route to studying lattice problems via quantum impurity models makes most sense for materials with strong local correlations. Such systems are often referred to as strongly correlated quantum materials, and their phenomenology often lies beyond conventional approaches like density functional theory or perturbative weak-coupling methods. In practise, such materials often involve transition metal elements or rare earth/actinide elements, due to the presence of partially filled $3d-$ or $4f-$orbitals in such elements. This is because these orbitals are more localised around the nucleus than the $s-$ or $p-$orbitals of comparable energy, and hence lead to a reduced itinerant energy scale as well as amplify the effects of local correlations [32].

The localised nature of these orbitals can be understood as follows. For the $3s-$orbital, the $1s$ and $2s$ orbitals have same orbital angular moment l but lower principal quantum number n; since all such wavefunctions must be orthogonal to each other, the $3s-$ orbital must develop nodes in its radial form and extend farther away from the nucleus in order to be orthogonal to these other orbitals. However, for the $3d-$orbital, there is no orbital with same l but lower n; this means that the orthogonality is already enforced by the behaviour of the angular part (the spherical harmonics), and the radial part can remain localised around the nucleus without having to spread out. A similar argument holds for the localised nature of the $4f-$ orbital.

We now discuss the phenomenology of some of these quantum materials in order to identify the challenges in trying to explain them and the kind of models that are likely to be useful in this endeavour.

Transition Metal Oxides (TMOs) Transition Metal Oxides (TMOs) are defined chemically by the presence of partially filled d-shells in transition metals (e.g., Cu^{2+}, Mn^{3+}, V^{4+}) coordinated by oxygen ligands. The physics of TMOs is governed by the competition between the kinetic energy hopping amplitude t and the on-site Coulomb repulsion U; due to the small overlap between the localised $d-$orbitals, the $d-$electrons can delocalise only by hopping onto the ligand atoms. According to the Zaanen-Sawatzky-Allen (ZSA) classification scheme [33], these materials are categorized as either Mott-Hubbard insulators, where the charge gap is determined by the d-d Coulomb interaction ($E_g \approx U$), or Charge-Transfer insulators, where the gap is dictated by the energy difference between the oxygen p-band and the metal d-band. These derive from considerations of several factors, such as crystal field splitting, arrangement of atoms and lattice symmetries.

For example, in the case of octahedral symmetry, repulsion from the surrounding atoms leads to the raising of energy of the $d_{x^2-y^2}$ and $d_{3z^2-r^2}$ orbitals forming the e_g doublet, while the other three orbitals (d_{xy}, d_{yz} and d_{zx}) form the t_{2g} multiplet. For the heavier TMOs (Ti, V, Cr...), the Fermi level lies within e_g. The increased positive nuclear charge (compared to the lighter elements) lowers the energy of the $d-$orbitals and brings them closer to the $p-$orbitals, reducing the charge transfer energy $\Delta = \varepsilon_d - \varepsilon_p$ of transporting an electron from a $d-$state to a $p-$state, compared to the $d-$level repulsion U ($|\Delta| < U$). Geometric reasons (the e_g orbitals point towards the $p-$orbitals) also enhance the hybridisation between $d-$ and $p-$orbitals in such materials. If Δ becomes larger than the bandwidth, we obtain a *charge-transfer insulator*, where the charge gap is decided by Δ. In the lighter TMOs, the Fermi level passes through t_{2g}, and because they point away from the $p-$orbitals, the hybridisation is weaker. Moreover, the reduced positive nuclear charge leads to an increased $\Delta > U$; a U greater than the bandwidth leads to what's called a *Mott-Hubbard insulator*.

Experimental studies of transition metal oxides (TMOs) have established metal–insulator transitions (MITs) as a defining consequence of strong electronic correlations in systems with partially filled (d)-electron shells [34, 35]. Early experiments on vanadates such as V_2O_3 characterised pressure-driven MITs through the pressure- and temperature-dependent conductivity as well as an anomalous enhancement of the optical conductivity [36–38]. Copper oxides constitute the most extensively studied class of correlated TMOs and provide a canonical example of a doping-driven MIT. Transport measurements on parent compounds such as La_2CuO_4 establish a robust insulating state, while carrier doping leads to anomalous metallic behavior characterized by linear-in-temperature resistivity and unconventional superconductivity [39–41]. Such metal-insulator transition is also reported from angle-resolved photoemission spectroscopy (ARPES) measurements in doped La_2CuO_4 and $Bi_2Sr_2CaCu_2O_8$, along with the emergence of pseudogap near the antinodes at high temperatures and a hole-to-electronlike Fermi surface reconstruction [42–44]. Nickelates ($LaNiO_2$, $La_3Ni_2O_7$, $La_4Ni_3O_{10}$, etc) exhibit related but distinct experimental signatures. With Ni^+ sharing the same $3d^9$ valence configuration as Cu^{2+} in the cuprate superconductors, these materials also display superconductivity, although the mechanism in the nicklelates is different due to the strong coupling between the layers [45, 46]. The optimal doped regime in thin films shows strange metal behaviour, while infinite-layer nickelates show the absence of long-range antiferromagnetic order [47].

Transition Metal Dichalcogenides Transition Metal Dichalcogenides (TMDs) are layered van der Waals materials with the stoichiometry MX_2, where M is a transition metal (e.g., Mo, W) and X is a chalcogen (e.g. S, Se, Te) [48,49]. Recent interest has focused on the interplay between strong spin-orbit coupling (SOC), electronic correlations and direct bandgap – properties that make them particularly useful for electronic applications, spintronics and DNA sequencing. Layered TMDs have the structure $X - M - X$, with the transition metal sandwiched between two hexagonal planes of chalcogenides. Different combinations of the metals with the chalcogens lead to widely varying structures and electronic properties in TMDs. Experimental studies show that metal–insulator transitions (MITs) in TMDs are commonly intertwined with charge density wave (CDW) formation rather than arising from purely on-site Coulomb localization [50, 51]. In prototypical compounds such as 1T-TaS$_2$, ARPES and transport measurements reveal fluctuating magnetic order parameters in the Mott state and a transition to superconductivity driven by pressure [52–54]. Scanning tunnelling microscope (STM) pulses have been successfully used to manipulate the metal–insulator transition of 1T-TaS2, demonstrating the macroscopic controllability of the Mott phase [55].

ARPES measurements in 2H-NbSe$_2$ and 2H-TaSe$_2$ exhibit partial gapping of the Fermi surface restricted to nested momentum regions and a temperature-dependent pseudogap similar to the cuprates [56, 57]. This is further supported by optical conductivity experiments that show spectral weight in the midinfrared region along with a low-frequency Drude contribution [58]. Transport measurements on single crystals of TaSe$_2$ reveal that disorder enhances superconductivity by suppressing other competing orders such as CDW [59]. Collectively, these experimental results establish TMDs as a distinct class of correlated materials in which reduced dimensionality, band narrowing, and lattice reconstruction cooperate to produce metal–insulator behavior beyond conventional band theory.

Iron-Based Superconductors The iron-based superconductors (FeSCs), discovered in 2008, constitute a broad family of layered materials that share a common structure: a layer of iron atoms joined by tetrahedrally coordinated pnictogen (P, As) or chalcogen (S, Se, Te) anions arranged in a stacked sequence; the layers are separated by alkali, alkaline-earth or rare-earth and oxygen/fluorine 'blocking layers' [60–63]. Doping or applying external pressure reveals coupled antiferromagnetic and structural transitions along with superconducting and nematic orders [64–66]. Electrical resistivity measurements in the neighbourhood of the quantum critical point show a crossover from a bad-metal behavior ($T-$linear resistivity) to a Fermi liquid behaviour (T^2-resistivity) [67]. Optical conductivity measurements indicate point to intermediate-to-strong electronic correlations [68].

ARPES measurements reveal a $k-$dependent superconducting gap and de Haas-van Alphen measurements indicate substantial quasiparticle mass enhancements, shedding more light on the correlated nature of the material [69, 70]. Strain-tuning measurements of the nematic susceptibility, and differential elastoresistance measurements have provided compelling evidence for the nematic transition and its surrounding fluctuations being the cause of the structural transition as well as the superconductivity [71, 72]. Neutron scattering and NMR measurements further reveal persistent low-energy spin fluctuations across wide regions of the phase diagram, supporting scenarios in which unconventional superconductivity emerges from the interplay of magnetic, orbital, and nematic degrees of freedom [73, 74].

Heavy-Fermion / Kondo-Lattice Systems Heavy-fermion materials are a class of strongly correlated electron systems typically realized in intermetallic compounds of rare earth metals (such as Ce, Sm and Yb) and actinides (such as U, Np and Pu), where localized f electrons hybridize with itinerant conduction bands [75, 76]. At high temperatures, these systems exhibit local-moment behavior governed by Curie–Weiss susceptibilities and incoherent transport, while at low temperatures the Kondo effect leads to the formation of a coherent many-body state with extremely large quasiparticle effective masses [77]. Experimentally, this crossover is most clearly manifested in transport and thermodynamic measurements: the electrical resistivity shows a characteristic maximum followed by a rapid drop signaling the onset of coherence, while the linear specific heat coefficient $\gamma = C/T$ reaches values orders of magnitude larger than in conventional metals, directly reflecting the heavy quasiparticle mass [78].

A defining feature of heavy-fermion materials is their proximity to magnetic quantum phase transitions, which can be accessed experimentally by pressure, chemical substitution, or magnetic field [79]. Near such quantum critical points (QCPs), transport measurements reveal pronounced non-Fermi-liquid behavior ($T-$linear resistivity) and a vanishing Hall response, along with a divergent effective quasiparticle mass as the transition is approached [80, 81]; these are all indications of the breakdown of electronic excitations near the transition. Neutron scattering experiments provide direct evidence for spin fluctuations down to very low temperatures, interplaying with the itinerant electrons at the QCP and potentially driving the superconducting transition [82, 83]. Quantum oscillation measurements through de Haas–van Alphen experiments further demonstrate Fermi surface reconstruction across the quantum critical point, highlighting the drastic change in Fermi volume [84]. Together, these experimental observations establish heavy-fermion compounds as paradigmatic systems in which strong local correlations, hybridization-driven coherence, and quantum critical fluctuations conspire to produce emergent low-energy quasiparticles and unconventional metallic behavior [85].

1.3.4 Generator of strong local correlations: Quantum impurity models

Within this thesis, we have developed an auxiliary model approach that combines the physics of quantum impurity models with a manybody version of Bloch's theorem to investigate interacting fermionic lattice models. This is motivated by the relative ease in solving impurity models and transparency afforded by such solutions. We now review the phenomenology of some of these impurity models and point out how the various impurity features affect the physics of the lattice models.

The single-impurity Anderson model describes the dynamics of a system of non-interacting itinerant electrons ($c_{k,\sigma}$) hybridising with a localised correlated impurity orbital (d_σ) [15]:

$$H_{\text{AIM}} = \sum_{k,\sigma} \epsilon_k n_{k,\sigma} + \sum_{k,\sigma} V_k (c^\dagger_{k,\sigma} d_\sigma + \text{h.c.}) + \epsilon_d \sum_\sigma n_{d\sigma} + U n_{d\uparrow} n_{d\downarrow} \ . \tag{3.18}$$

V_k controls the hybridisation strength, and ϵ_d and U control the onsite energy and local correlation respectively of the impurity site. In the limit of large U, the impurity dynamics gets projected onto its spin states, and the model then describes the scattering of non-interacting electrons off magnetic impurities, through the Kondo model [86]. Formally, the Kondo model is obtained by carrying out a Schrieffer-Wolff transformation on the Anderson impurity model and then using the condition to $V \ll U$ to drop all contributions beyond second order in the ratio V^2/U [87]:

$$H_{\text{Kondo}} = \sum_{k,\sigma} \epsilon_k n_{k,\sigma} + J \sum_{k_1,k_2} \sum_{\alpha,\beta} \vec{S}_d \cdot \vec{\sigma}_{\alpha,\beta} c^\dagger_{k_1,\alpha} c_{k_2,\beta} \ , \tag{3.19}$$

where $\vec{S}_d$ is the impurity spin formed by projecting the Hilbert space of the impurity to the singly-occupied states.

Even though the conduction bath is non-interacting, the presence of local repulsion on the impurity site makes the problem *many-electron* in nature: that is, the "U" term makes it impossible to study the electrons in isolation. This can be seen as follows: imagine two conduction electrons $|k_1 \uparrow\rangle$ and $|k_2 \downarrow\rangle$ trying to hybridise with an empty impurity site. The first electron moves onto the impurity site through a scattering process of the form $c^\dagger_{d\uparrow} c_{k_1\uparrow}$ and makes the impurity site single occupied ($|\uparrow\rangle_d$), but now the second electron k_2 *must pay a cost* of the order of U in order to hybridise with this singly-occupied impurity site through a process of the kind $c^\dagger_{d\downarrow} c_{k_1\downarrow}$. This shows that the conduction electrons *are aware* of the other electrons, and all of them must be treated together. Another way of seeing this collective behaviour is from the Kondo model. Again imagine two spin-up electrons k_1, k_2 that are about to scatter off the impurity spin (which is presently down) via a *spin-flip* process. The first electron scatters and leaves the impurity spin up while it itself becomes down, but now the second conduction electron cannot spin-flip scatter because the spins cannot be exchanged (both are down!). This is another example of how the conduction electrons can *feel* the presence of the other electrons, and this is what makes the problem many-electron in nature.

Kondo screening Despite the interacting nature of the problem, the problem can be solved in an effectively exact fashion, via Wilson's numerical renormalisation group approach [88, 89], density-matrix renormalisation group approach [90] or the method of *Bethe ansatz* [91, 92]. The former solution is more transparent: for any non-zero value of the hybridisation V (or Kondo coupling J), the low energy physics of the impurity is described by a strong-coupling fixed point ($V = \infty$, or $J = \infty$), where the impurity forms a maximally-entangled singlet state with the local spin density of the conduction bath. The low-energy excitations about such a fixed point can be calculated from renormalised perturbation theory, and describe a local Fermi liquid – one particle excitations in the neighbourhood of the singlet renormalised by a local repulsion term [93, 94]. The crossover from the free local moment at high temperatures to the screened non-magnetic state at low temperatures occurs at an emergent energy scale T_K, the Kondo temperature, which depends exponentially on the strength of the Kondo coupling, and is the only energy scale remaining at low temperatures, so that all thermodynamic quantities must go as a function of T/T_K [88].

At half-filling (attained for $\epsilon_d = -U/2$), the only other fixed points are the free orbital fixed point at $V = 0, U = \epsilon_d = 0$ where the impurity site is freely occupied by two independent spins, and the local moment fixed point at $V = 0, U = \infty$, where the impurity site is occupied by a local moment that is decoupled from the bath [77, 89]. Away from half-filling, two other fixed points become available [77, 89]: a valence fluctuation fixed point at $V = 0, \epsilon_d = 0, U = \infty$ where the doubly-occupied state has been projected out from the symmetric free orbital fixed point, and a frozen impurity fixed point at $V = U = 0, \epsilon_d = \infty$ where the impurity site is left completely empty.

The crossover from local moment physics to Kondo screening as the system is tuned from high to low temperatures can also be seen from the impurity spectral function [95–98]. At low frequencies, the spectral function exhibits a sharp resonance around $\omega = 0$ that indicates the presence of gapless single-particle local Fermi liquid excitations. At higher frequencies, one sees satellite peaks that involve impurity excitations that cost energy of the order of U. The height of the central Kondo resonance is a topological invariant; it is fixed by the impurity occupancy owing to an application of the Friedel sum rule to impurity models [99], and remains unchanged as the interaction U is increased adiabatically while remaining at particle-hole symmetry. The topological nature can also be ascertained by identifying such a sum rule as the impurity model analogue of Luttinger's theorem [24, 25] that relates the total number density of electrons to the number of Greens function poles within the Fermi volume, and which is itself topological [100, 101].

Breakdown of Kondo screening While the vanilla Anderson model (or the Kondo model derived from it) only display a strongly-coupled screened phase, it is possible to tweak it in order to introduce Kondo frustration which can eventually lead to a breakdown in the Kondo screening process. Such models are interesting because they provide a means of studying zero temperature *continuous* phase transitions and address experimental signatures of quantum criticality in correlated systems. The canonical example of this is the multichannel Kondo model, involving a single impurity hybridising with multiple independent conduction channels [102]:

$$H_{\text{MCK}} = \sum_{k,\sigma,\alpha} \epsilon_{k,\lambda} n_{k,\sigma,\lambda} + \sum_{k_1,k_2,\lambda} J_\lambda \sum_{\alpha,\beta} \vec{S}_d \cdot \vec{\sigma}_{\alpha,\beta} c^\dagger_{k_1,\alpha,\lambda} c_{k_2,\beta,\lambda} , \tag{3.20}$$

here λ is the channel index. In the case of an isotropic Kondo coupling across the channels ($J_\lambda = J$), the strong-coupling $J = \infty$ fixed-point of the single-channel Kondo model is replaced by an intermediate-coupling fixed-point at a finite value J^* [102, 103]. Powerful methods like boundary conformal field theory [104–109], bosonization [110–114], numerical renormalization group [106, 115, 116] and Bethe ansatz [117–121] have shown that the low energy theory of such models exhibit non-Fermi liquid excitations. Further, the system displays anomalous behaviour in thermodynamic properties near $T = 0$ and a fractional zero temperature entropy in the thermodynamic limit (upon ensuring that temperature is taken to zero after the system size is taken to infinity [113, 122]). This anomalous divergent behaviour is a signature of the fact that the MCK system with a single channel-symmetric exchange coupling is quantum critical: the ground state is susceptible to perturbations that introduce channel anisotropy in the exchange couplings [102, 106, 112, 120, 123].

Another class of models that shows Kondo breakdown is that of pseudogap Anderson and Kondo models, where the impurity hybridises with a conduction bath whose density of states (DOS) varies as $\rho(\omega) \sim |\omega|^r$ with a paramter $r > 0$ that controls how fast the DOS vanishes as the Fermi level is approached. Such behaviour can arise in semi-metals and in long-range ordered systems where the order parameter vanishes at certain points on the Fermi surface (such as $d-$wave superconductors). The pseudogap impurity models exhibit a boundary quantum phase transition between a Kondo-screened strong-coupling phase and an unscreened local-moment phase as a function of interaction strength or exchange coupling [124–127]. At the critical Kondo coupling, the impurity spectral function and spin susceptibility exhibit nontrivial power-law scaling with anomalous exponents determined by r [125, 128–131]. For $0 < r < 1/2$ (in the particle-hole symmetric case), both the local-moment and strong-coupling phases are stable, separated by a critical fixed point characterized by fractional residual entropy and nontrivial hyperscaling relations [125, 132]. For larger exponents, the strong-coupling phase may disappear entirely, and the system remains in the local-moment phase for all couplings [127].

The multi-impurity Anderson and Kondo models furnish another canonical realization of Kondo breakdown, driven by the competition between Kondo screening and inter-impurity interaction. The paradigmatic example is the two-impurity Kondo model (TIKM), in which two localized spins couple both to conduction electrons and to each other via an antiferromagnetic exchange K [133, 134]. In the limit of large $K > 0$, the impurities form a non-magnetic singlet decoupled from the conduction electrons and suppressing the Kondo resonance. This fixed point is separated from the strong-coupling Kondo screened fixed point through a quantum critical point [133–135] displaying logarithmic or power-law scaling in thermodynamic quantities and fractional residual entropy [136]. A modified version of this model that uses conduction bath-mediated inter-impurity interaction leads to a metallic spin liquid displaying nonlocal impurity-spin entanglement and fractionalized charge excitations [137]. More intricate geometries can be constructed by arranging antiferromagnetically coupled Kondo impurities along a triangle, each hybridising with its own conduction bath; such a model displays a topological phase transition from a symmetry-preserved gapless spin liquid phase with ”topological order” to a screened Fermi liquid phase [138]. Indeed, recent experiments have validated such phase diagrams for the case of two impurities by studying pairs of copper atoms in the presence of strong magnetic fields [139].

1.4 Phenomenology of Mott transition in lattice models

1.4.1 The Infinite-Dimensional Hubbard Model

The Hubbard model (introduced in eq. 1.1) presents one of the most fruitful playgrounds for investigating correlation-driven metal-insulator transitions and its various facets. This model, despite its conceptual simplicity, exhibits a rich variety of phenomena, including antiferromagnetism, superconductivity, and, most notably for our purposes, the correlation-driven Mott metal–insulator transition (MIT) [4, 140–142]. While exact solutions exist in the limits of $d = 1$ [143] and $d = \infty$ [144], the problem remains largely open in more realistic situations such as two and three spatial dimensions.

As discussed earlier, the limit of infinite dimensions presents a drastic simplification for the manybody scattering processes within the problem: the one-particle $k-$space self-energy becomes momentum-independent due to the vanishing of all non-local contributions (they scale as negative powers of the coordination number, which diverges in the limit of $d = \infty$). This simplification lies at the heart of dynamical mean-field theory and its ability to capture the Mott metal–insulator transition at both zero and finite temperatures for this limiting case [18, 144–148]. At half-filling and zero temperature, DMFT predicts a continuous transition as a function of U wherein the quasiparticle weight Z tends to zero at a critical interaction strength U_{c2}, signaling the vanishing of coherent Fermi liquid behavior and the opening of a Mott gap in the single-particle spectral function [18, 146]. The three-peak structure of the interacting spectral function, with a central quasiparticle resonance flanked by incoherent lower and upper Hubbard bands, indicates the presence of highly renormalised Landau quasiparticles [146]. A doping-driven continuous metal-insulator transition also exists for this model at zero temperature, marked by the vanishing of electronic compressibility at a critical value of the chemical potential [149]. The quantum critical point itself (at $T = 0$) is likely to be an exotic state, and important questions about it remain unanswered. *Is it possible to obtain a low-energy theory for the local gapless excitations precisely at the MIT, where the metal is on the brink of destruction? How do these excitations compare with those of the local Fermi liquid, e.g., in terms of self-energies and two-particle correlation functions?*

While the DMFT solution captures the zero temperature transition accurately, the self-consistently determined nature of the conduction bath renders the solution somewhat opaque and makes it difficult to write down an effective Hamiltonian for the low-energy theories. Since the vanilla Anderson impurity model does not contain the mechanism to stabilise a localised phase over a finite-sized parameter regime, an important question arising from the DMFT solution is: *what are the minimal changes that one must implement into the Anderson impurity model in order to capture such a metal-insulator transition ($1/2$-filled Hubbard model on the Bethe lattice in $d = \infty$), in sufficient accuracy and precision?* Finding such an impurity model would also reduce the considerable computational effort that is presently required in self-consistent approaches.

At finite temperatures, the interaction-driven as well as doping-driven Mott transition become first order, and is characterized by a region of phase coexistence and hysteresis [147, 150]. Two spinodal lines, $U_{c1}(T)$ and $U_{c2}(T)$, emerge; within this region metallic and insulating DMFT solutions coexist, and responses such as resistivity show a significant discontinuity [151]. These lines meet at a finite-temperature critical endpoint (U_c, T_c), above which there is no sharp distinction between metallic and insulating solutions and only a crossover exists between the Mott insulator and the Fermi liquid through a bad-metal regime, as the chemical potential is tuned [147, 149, 152–154]. The coexistence of metallic and insulating phases shows that the insulating solution is present within the many-body spectrum of the metallic phase. This naturally leads to the following question: *since a quantum impurity lies at the heart of the self-consistency procedure, what does the presence of coexistence region mean for the Hamiltonian dynamics and the spectrum of this impurity model? Can the impurity model Hamiltonian display the emergence of the insulating state prior to the transition, through changes in the eigenstates?* Analytic insight of this kind is particularly important because approaching a coexistence region numerically is often fairly tricky.

The bad metal regime displays anomalous linear-in-temperature resistivity which corresponds to quantum critical ω/T scaling, along the quantum widom line [155, 156]. This is accompanied by the disappearance of the Drude peak in optical conductivity, signalling the loss of quasiparticle transport [155]. Similar quantum critical scaling is also obtained from DMFT solutions at very low temperatures, hinting at a "hidden" quantum criticality that is continuously connected to and responsible for the critical behaviour above the finite temperature endpoint [157]. Luttinger's theorem, which relates the poles of the Greens function within the Fermi volume to the number of electrons in the system at zero temperature, is satisfied within the Fermi liquid phase but is violated in the particle-hole asymmetric Mott insulator by a universal magnitude [158]. The emergence of a quantum critical regime at high temperatures and its connection to similar signatures at low temperatures suggests the presence of quantum fluctuations in the underlying impurity model that modify the impurity-bath scattering processes and the associated impurity responses. One can therefore ask: *What are the fluctuations that destroy the metal and lead to the insulating phase? Can we obtain a universal theory for these competing tendencies?*

The above studies of single band models describe the Mott transition to be of the Mott-Hubbard kind (described in previous sections), where the size of the gap is determined by the double-occupancy cost U. In order to obtain transition of the charge-transfer kind, one needs to consider at least a two-band model. Indeed, in a non-degenerate two-band Hubbard model with a charge-transfer energy Δ for hopping from a $d-$orbital to a $p-$orbital, one sees a Mott-Hubbard transition when $U < \Delta$ and a charge-transfer transition when $U > \Delta$ [159]. The degenerate two-band model with inter-band local correlation shows successive Mott-Hubbard transitions at low temperatures whenever the filling passes through integer values and the interaction strength is intermediate [160].

1.4.2 The two-dimensional Hubbard model

The 2D Hubbard model is one of the most well-studied models in condensed matter physics, particularly because of its paradigmatic role in Mott metal-insulator transition and its connections to the physics of copper oxide materials and $d-$wave superconductivity. Photoemission experiments reveal the presence of p and d bands close to the Fermi surface in such materials [42]; further work by Zhang and Rice showed that a specific combination of these orbitals governs the low-energy physics, leading to an effective description in terms of a single band Hubbard model [161]. The discovery of high-temperature superconductivity in nickelates and organic charge transfer salts has provided more platforms where the Hubbard model and its extensions can prove to be useful [162, 163].

While simple in appearance, the model has proved to be notoriously hard to solve except in highly specific parameter regimes, despite several decades of computational effort. This is due to several reasons [142]: (i) the presence of closely-lying energy scales arising from competing tendencies, (ii) the inability of several methods to either reach zero temperature or extrapolate to the thermodynamic limit, (iii) the tendency of approximate/finite-size methods methods to overestimate proclivities towards spontaneous symmetry breaking, and (iv) the presence of differing phases separated by slow crossovers without any sharp boundaries.

At half-filling, the 2D Hubbard model has a spectral gap to charge excitations for any non-zero interaction strength [164–168]. The behaviour below the charge gap is described by a nonlinear sigma model with parameters that depend on the interaction strength U [169]. Upon tuning to finite temperatures, the system goes through a sequence of crossovers from the insulating ordered state to a Fermi liquid state with coherent quasiparticles into a non-Fermi liquid state with partial gapping of the Fermi surface [166, 168, 170]. *Capturing the momentum dependence of static and dynamical quantities in sufficient detail remains an open challenge in such problems, since results from cluster methods suffer from sensitivity to periodisation schemes [171].*

Away from half-filling but at weak coupling, the Neel order is accompanied by incommensurate magnetic orders (also called spiral magnetic state) with wave vectors away from (π, π), as seen from mean-field studies [172, 173]. Functional RG studies indicate that such solutions can exist up to fairly high doping if the superconducting instability is suppressed [174, 175]. Superconductivity with $d-$wave pairing has been achieved from renormalised perturbation studies as well as from diagrammatic Monte Carlo (diagMC) and functional RG (fRG) calculations [176–178]. More intricate competition between the superconducting and stripe states is found in three-band Hubbard models from variational Monte Carlo, cluster DMFT and DMRG calculations [179–181]. *Further work is necessary to resolve the closely-lying phases in such models.*

The phenomena of pseudogap (PG), where the Fermi surface acquires a momentum-dependent suppression of spectral weight, has also been observed above a certain temperature from methods like diagMC and fRG [168, 182]. The gap is seen to open first near the antinodal point $(\pi, 0)$ and spread across the Fermi surface until it reaches the node $(\pi/2, \pi/2)$. The mechanism of the pseudogap at weak to intermediate coupling is found to be long-range antiferromagnetic (AF) correlations [183]. The pseudogap disappears at large doping when the Fermi surface undergoes a topological transition from hole-like to electron-like. At strong coupling, the competition between various phases such as $d-$wave superconductivity, antiferromagnetic order and stripe states (spin/charge order modulating in real space) is an active subject of study. Recent studies using various methods like DMRG and iPEPS found that the period 8 charge stripe state is lower in energy than the superconducting state at strong coupling $U/t = 8$ and no next-nearest-neighbour hopping [184]. This was later corroborated by studies from variational Monte Carlo and other DMRG studies [185, 186]. *It is not yet clear whether the stripe period gets lowered as the interaction strength is increased.* At weaker coupling, the above methods report a reduced stability for the strip state and increased tendency to realise a $d-$wave SC state [177, 187]. *Further investigation is necessary to map the crossovers between stripe and superconducting phases as doping is tuned. Moreover, the precise parameter regime in which the $d-$wave superconducting state becomes more stable than the stripe states is still an open question.*

At strong-coupling, the pseudogap is obtained below optimal doping and has been studied using dynamical methods such as cluster DMFT (DCA, dynamical cluster approximation) and cellular

DMFT (CDMFT) [188–191]; in contrast to the weak-coupling feature (where the PG was borne from long-range AF fluctuations), it is believed that the momentum-space differentiation in this regime is brought about by the proximity to the Mott insulator [190]. Conflicting indications are however provided by a fluctuation analysis that finds that long-lived long-ranged spin-fluctuations at the antinode are responsible for the opening of the PG [192]. The PG is found to be accompanied by non-Fermi liquid behaviour in the one-particle self-energy [189]. The doping value at which the non-Fermi liquid first appears was found to be lower for electron-doping than hole-doping [193]. At half-filling, DCA calculations reveal that the extent of the PG decreases as the next-nearest-neighbour hopping strength is increased [194]. Moreover, while DCA+continuous-time auxiliary field methods do not show any effect of van Hove singularities on the generation of the PG [194], DCA+determinantal Monte Carlo approaches identify the proximity to van Hoves as favouring the formation of a PG [195]. *Methods with better momentum-space resolution is necessary in order to identify the role of nesting and van Hoves on the pseudogap.*

The question of whether superconductivity can arise from repulsion was one of the primary motivations for studying the Hubbard model. DiagMC calculations reveal superconductivity in the ground state at weak-coupling and moderate doping, and with multiple symmetries of the gap [177, 196]. For higher couplings (which is relevant to the cuprates) and moderate doping, stripe states take over. Quantum Monte Carlo and DCA calculations suggest a $d_{x^2-y^2}$ pairing symmetry [165, 197]. DCA calculations on the 2D Hubbard mode suggest low-frequency spin fluctuations as the pairing glue for the superconductivity [198]. *The description of how various parent phases such as the pseudogap and strange metal precipitate superconductivity, in terms of the nature of fluctuations and the appropriate degrees of freedom, is still an open question..* Apart from superconductivity, several strongly correlated materials also exhibit strange metallicity, with transport properties that defy Fermi liquid theory, such as a linear-in-temperature resistivity that extends across large regimes of temperature. Within the Hubbard model, evidence for such $T-$linear resistivity and robustness across temperatures is available from CDMFT and determinantal Monte Carlo calculations [199, 200]. *More reliable methods are needed to access and understand the intermediate temperature regime of strange metals [142, 201].*

1.4.3 Theories of heavy fermion materials

Heavy fermion materials represent the canonical environment for studying quantum phase transitions in correlated systems, owing to the explicit observation of quantum critical points at the boundary of antiferromagnetic order [202]. These systems involve the coexistence of magnetic moments (arising from localised $4f-$ and $5f-$electrons) and itinerant conduction electrons. The ground state of such a system is dictated by the competition between two kinds of interactions: internal spin-spin interaction J_{RKKY} between localised moments in the $f-$layer, and interaction J_K between $f-$electrons and conduction electrons [203].

Analogous to the screening of a single moment by a conduction bath, the lattice of $f-$moments are prone to undergoing Kondo screening by the conduction electrons through the inter-orbital hybridisation J_K. The local Fermi liquid excitations associated with the spins overlap with each other and become delocalised, leading to a paramagnetic heavy Fermi liquid with highly renormalised quasiparticle mass [204]. Because of the delocalisation of the excitations of the $f-$spin through

Kondo screening, the number of carriers increases to $1 + x$ compared to x coming purely from the conduction electrons; this is commonly referred to as the large Fermi surface phase [205]. Indeed, a perturbation theoretic calculation in the on-site interaction strength of the $f-$layer allows one to rewrite the two-orbital model as two decoupled quasiparticle bands with dispersions renormalised by phenomenological parameters like the quasiparticle residue [77]. These phenomenological parameters can be estimated from mean-field approaches [206, 207]. While the temperature T^* at which coherent Landau quasiparticles can proliferate was found to be generically lower than the single impurity Kondo temperature T_K [208], T^* was found to be higher than T_K near half-filling of the conduction layer [209]. At half-filling in both layers, the chemical potential passes through the middle of the gap between the two quasiparticle bands, and the system is a Kondo insulator [210]; for bipartite lattices, the ground state can be generally shown to be unique and a singlet ($S = 0$) [211–213].

Tuning the $f-$layer J_{RKKY} to large values compared to the inter-layer hybridisation can bring forth a quantum phase transition into a different ground state [214]. Experimental evidence suggests that this transition usually falls into one of two classes, depending on the behaviour of the Kondo singlet binding energy E^* in the paramagnetic phase. If E^* remains non-zero as the QCP is approached from the heavy Fermi liquid into the ordered state, the transition is said to fall in the SDW type, because it can be described as an instability of the heavy quasiparticles into a spin-density-wave ordered phase [215–217], in the form of a quantum action that involves the spatial and temporal fluctuations of an order parameter. $CeCu_2Si_2$ is a prototypical material that exhibits quantum criticality of the SDW type [76]. Because of the non-zero binding energy of the singlet state at the QCP, heavy quasiparticles survive across the transition and the Fermi surface is expected to evolve continuously, along with the order parameter [218, 219].

On the other hand, if E^* vanishes at the QCP, the transition belongs to the Kondo breakdown type, which is not compatible with the spin-density-wave scenario [220]. This happens when gapless spin fluctuations within the $f-$layer compete with the Kondo spin-flip processes and lead to the destruction of Kondo screening on the Fermi surface [79]. This is tracked by the vanishing of the quasiparticle residue of the heavy Fermi liquid as one approaches the QCP from the paramagnetic phase [131, 219]; this in also implies the critical non-Fermi liquid nature of the large Fermi surface at the transition [221]. *A complete understanding of the Kondo breakdown class of QCP is still missing in the literature.*

Due to the destruction of Kondo screening across the QCP in this class of materials, the localised $f-$spins cease to contribute to the Fermi surface away from the paramagnetic phase, and the Fermi surface shrinks discontinuously from large $(1 + x)$ to small (x) across the QCP [222]. *Such a change in the Fermi volume violates Luttinger's theorem which equates the particles density to the Fermi surface volume, and Oshikawa's topological flux-tuning argument suggests the presence of additional degrees of freedom that respond to the flux while not being part of the Fermi surface. Additional work is needed to identify these degrees of freedom.*Materials like $YbRh_2Si_2$ and $CuCu_{6-x}Au_x$, among others, appear to exhibit behaviour that falls in this class [223, 224]. As mentioned before, several features of this class of experiments are not compatible with the SDW instability transition of the heavy Fermi liquid; these include the jump discontinuous change in Fermi volume mentioned previously (and an associated discontinuous change in the hall conductivity) [225], vanishing of the scale T_{FL} (associated with the onset of Fermi liquid behaviour) and the divergence of the quasiparticle mass upon approaching the QCP from the paramagnet [220], a universal linear-in-temperature strange metal behaviour of the resistivity in an intermediate finite temperature quantum critical regime above the QCP [223, 226] and the possible lack of magnetic ordering immediately away from the QCP [227, 228]. *At present, the connection of the physics of heavy fermion models like the periodic Anderson model to strange metal behaviour is an open question; the mechanism for generating such behaviour from the QCP remains to be completely understood.*

A fairly comprehensive model for describing the phase transition in heavy fermion systems is the periodic Anderson model (PAM):

$$H_{\text{PAM}} = \varepsilon_f \sum_{i,\sigma} n_{i,f,\sigma} + U_f \sum_{i} n_{i,f,\uparrow} n_{i,f,\downarrow} + \sum_{k,\sigma} \epsilon_k n_{k,c,\sigma} + V_\perp \sum_{i,\sigma} \left(c^\dagger_{i,f,\sigma} c_{i,c,\sigma} + \text{h.c.} \right) , \tag{4.1}$$

where the subscripts f and c in the operators indicate whether they belong to the $f-$layer or the conduction electron layer. At the Kondo breakdown QCP and in the RKKY-dominated phase, the inter-layer hybridisation is expected to be irrelevant and the $f-$ and conduction electrons fail to bind into a Kondo singlet in the ground state [229, 230]. Several numerical and analytical methods have been applied to investigate various parts of the phase diagram and probe varying aspects of the phenomenology. *However, many of these methods suffer from a lack of predictions of various transport measures like strange metallicity, and they often suffer from insufficient resolution in frequency or momentum space.*

One of the major approaches applied to this problem is that of extended DMFT (EMDFT), in which one maps the PAM to a self-consistently determined Bose-Fermi-Kondo model [131, 231]. One of the smoking gun signatures of quantum criticality is the scaling of dynamical properties as $\omega/k_B T$ due to temperature being the only remaining energy scale in the system; this is captured from EMDFT calculations [219]. Numerically exact unbiased methods like DMRG and auxiliary field quantum Monte Carlo demonstrate the abrupt change in Fermi surface volume across the transition [232, 233]. Parton methods that implement a slave-particle representation followed by a $1/N$ expansion reveal jump in the conductivity across the transition, with the small Fermi surface arising from a spin-charge separation of the $f-$electrons into a fractionalised Fermi liquid phase [234, 235]. *Currently, a complete description of the Kondo breakdown driven transition that captures the behaviour of both the heavy Fermi liquid and the RKKY-dominated phase is missing.*

1.5 An alternative perspective on metal-insulator transitions: Entanglement renormalisation and holography

Role of entanglement in fermionic systems The present millennium has seen a rapid surge in the use of quantum entanglement towards the study of quantum and condensed matter systems [236–241]. Being non-local by nature, it is helpful in tracking correlations among particles across both small and large distances, giving valuable insight into the nature of the ground state and the low-energy excitations. Entanglement, in the form of various measures like entanglement entropy and mutual information, is the engine that gives rise to new low-energy degrees of freedom from a renormalisation group (RG) perspective, and is therefore key to understanding and characterising phase transitions and criticality in quantum matter. For instance, entanglement entropy is at the heart of understanding the so-called topological states of matter that can be described in terms of a purely topological effective Lagrangian and have a robust ground state degeneracy and gapless edge states [242–262]. In contrast to systems with local interactions whose subsystem entanglement entropy scales with the area of the subsystem [238, 263, 264], topologically ordered systems have an additional subdominant topological term dubbed the topological entanglement entropy [265–283] that quantifies the long-ranged nature of the entanglement in such systems. The non-local nature arises from the fact that the ground state of such a system cannot be made separable (non-entangled) by the application of purely local unitary transformations [240, 284, 285].

More pertinent to the work at hand, however, is the entanglement in free fermionic systems. In this thesis, we will consider relativistic Dirac fermions in two dimensions, both massless and massive. They emerge in several condensed matter systems [286], appearing at the surfaces of three-dimensional topological insulators [287–290], quasi-2D organic materials [291–293] and in arficial materials like cold atoms, molecular and microwave crystals [294–300]. For massless critical systems (or conformal field theories (CFTs)) living in one spatial dimension, the entanglement entropy of a single subsystem of length l scales as $\sim \frac{c}{3}\ln l$ [301–310], where c is the conformal central charge of the CFT. The case of multiple subsystems is more involved, but has been solved [308, 311, 312]. For massive fermions, the entanglement entropy (or rather the entropic $c-$function, which is the derivative of the entropy along the distance) can be expressed in the form of differential equations which need to be solved numerically [311, 313–315]. The solutions lend themselves to small and large mass expansions, reducing to $\frac{1}{3}\ln\frac{l}{\epsilon} - \frac{1}{6}\left(ml\ln ml\right)^2 + O((ml)^2)$ for the case of small ml, and to $\sim (ml)^{-1/2}e^{-2ml}$ for that of large ml, both for a single interval. The $\ln l$ behaviour is a specific case of the more general $l^{d-1}\ln l$ violation of the area law, as observed in $d-$dimensional fermionic critical systems [311, 316–330]. This violation arises from the non-local nature of the quantum fluctuations that exist in quantum critical fermionic systems.

While there is no geometry-independent subdominant term in the entanglement entropy of free fermionic systems [331, 332], it *is* possible to generate such a term by placing the system on a manifold with a non-trivial topology, say a cylinder, and inserting a magnetic flux through the cylinder [331, 333–336]. For a massless Dirac fermion, if there are ϕ number of electronic flux quanta piercing the system, the additional term that is generated in the entanglement entropy of a subsystem of sufficiently (as defined in Ref. [331]) large length is of the form $\frac{1}{6}\ln\left|2\sin\frac{\phi}{2}\right|$ [331, 334]. There is evidence from numerical computations on discrete models that such a term is a signature of the specific field theory and encodes universality [336–338]. In this thesis, we put this idea on a stronger analytical footing by relating the flux-dependent term to a topological invariant of the field theory, its Luttinger volume [24, 25, 100, 101, 339–341]. Given by the number of $k-$states enclosed by the Fermi surface, Luttinger volume V_L has proved to be one of the most important indicators that distinguish metals from insulating states of matter. In metals, V_L is constrained by the number of particles in the system, following Luttinger's theorem [24, 25, 100]. As shown most prominently in Ref. [100], Luttinger's theorem depends crucially on the presence of extended states in the metal that can respond to flux insertion experiments. We demonstrate that indeed it is this subdominant term in the entanglement entropy that carries this dependence and hence encodes Luttinger's theorem. It is known that for strongly-interacting states like Mott insulators or quantum hall states, V_L is modified due to changes in the topology of the Fermi surface [101, 339–341]. In this context, we relate the topological invariant on the metallic side (Luttinger volume) to the topological invariants (the Chern numbers) in the quantum hall system that the metal would lead to when placed in a perpendicular magnetic field. Such a relation makes explicit the transmutation of the extended states into the localised states of the insulating phase.

Entanglement hierarchy and the holographic principle The leading geometry-dependent piece is also found to have intriguing properties of its own. The fact that the free fermion Lagrangian is diagonal in momentum space means that the entanglement arises purely from real space quantum fluctuations, and this allows us to study the entanglement of subsystems of varying sizes in momentum space. We find that the entanglement entropy hence obtained has a Russian doll-like nested structure, such that increasing the size of the subsystem in momentum space gradually reveals layers of entanglement. In fact, we argue that such transformations between the subsystems amount to a renormalisation group (RG) flow in the space of the field theories with varying spectral gaps, and the above-mentioned hierarchy in the entanglement entropy then stretches across the RG transformations. The precise prescription for choosing the subsystems is described in Sec. **??**. We also show, simultaneously, that the hierarchy holds across multipartite entanglement measures, leading to an onion-like structure where increasing the number of parties in the entanglement measure (mutual information, tripartite information, 4-partite information and so on) leads to additional entanglement that did not exist before. More importantly, we interpret the hierarchy of entanglement along the RG flow as the emergence of an additional dimension [342–344], allowing us to define a measure of distance using some non-negative measure of entanglement, like the mutual information [345–348]: the more the entanglement, the less will be the distance between those points. Generation of an additional dimension is, of course, the essence of the AdS/CFT conjecture or, more generally, the holographic principle.

The holographic principle [349–357], also known as the gauge-gravity duality, posits that there exists a duality relation between a quantum field theory with a gauge field, and a gravity theory having one additional spatial dimension [358–361]. The additional dimension can be visualised as a renormalisation group flow that starts from the boundary field theory and extends into the bulk [362–371]. The most popular realisation of the holographic principle is the AdS/CFT correspondence that links a super-symmetric gauge theory with a string theory [372–374]. The essence of holography is that all the information about the bulk gravity theory is already contained within the boundary gauge theory; this idea is very similar to the discovery by Bekenstein and Hawking [375–377] that the entropy of a black hole is determined by its surface area. In fact, Ryu and Takayanagi proposed a generalisation of this result where the entanglement entropy of subsystem region a CFT is determined by the area of the minimal surface that bounds the region in the bulk [378, 379]. The AdS/CFT correspondence happens to be a strong-weak duality - it relates a strongly coupled CFT to a theory of weak classical gravity. This allows physicists to use the holographic principle and convert a problem of strong quantum correlation to one of classical gravity which can be solved using well-known methods [352, 353]. This has led to new insights in various topics of high energy and condensed matter physics, such as the viscosity of the quark-gluon plasma [380], the response functions of the Bose-Hubbard model [381, 382], and the phenomenology of non-Fermi liquids [383–386], holographic superconductors [387–392], striped phases [393–395] and holographic Fermi liquids [396–400]. An overview of the metallic states (phases that satisfy Luttinger's theorem) and their connections to the gauge-gravity duality have been provided in Refs. [401–403].

Of course, on the flip side, the holographic principle also implies that studying the boundary conformal theory can provide insights into the nature of the bulk quantum gravity theory. This however requires constructing, *ab initio*, the emergent dimension by starting from the field theory, and has proved to be more difficult than the converse, mostly because the field theories are often strongly coupled. Nevertheless, there exist some examples of constructive attempts towards a bulk gravity theory [340, 341, 347, 348, 404–437]. Some well-known approaches include the momentum shell renormalisation approach of Sung-Sik Lee [417–420] and the multi-scale entanglement renormalisation ansatz (MERA) of Vidal [421–425]; both use certain forms of RG flow to generate the extra spatial dimension. The former integrates out high momentum degrees of freedom in favour of generating dynamics for the boundary gauge field, with the RG trajectory variable giving rise to the additional spatial dimension [351]. On the other hand, the MERA works by disentangling degrees of freedom in real space, creating a tensor network in the process [426–428] and leading to a renormalisation in the entanglement. The renormalised entanglement is interpreted as the additional dimension [423]. There exist other approaches that also proceed via entanglement renormalisation, such as the exact holographic mapping [347, 348] (EHM) and the unitary renormalisation group [340, 341, 429] (URG) that operate purely through unitary transformations and have been explicitly shown to lead to a holographic tensor network along the RG flow. Apart from these, there is a body of work due largely to Ki-Seok Kim that provides a geometric description of Wilson's numerical renormalisation group [89,430–432,438], and creates a more general RG framework in terms of an order-parameter field that extends the Ads/CFT correspondence to non-critical theories [433–437].

Questions on entanglement and holography become particularly interesting in systems of critical fermions. As mentioned earlier, such systems differ from bosonic and gapped systems through the presence of longer-ranged entanglement that follows a modified area-law [322, 324] as well as topological features arising from the incompressibility of the Fermi volume [100, 101]. Fermi surface gapping phase transitions in such systems are often driven by changes in topological quantum numbers and the nature of the inter-particle entanglement [79, 340, 341]. Constructing a framework for fermionic criticality is therefore an active area of research with consequences for important systems such as unconventional superconductors, strange metals and heavy fermions [80, 439–441]. Still less is known about the holographic implications of such systems. We extend this body of work by providing an explicit demonstration of the holographic principle in the form of a simple and tractable exact holographic mapping (EHM) for a free fermion system with and without a mass gap. Our program yields analytic relations between the RG flow parameters and their dual bulk quantities such as distances and curvature. This also allows us to study the consequences of a critical Fermi surface (lying precisely at the quantum phase transition) on the nature of the holographic space.

1.6 A New Auxiliary Model Approach Towards Fermionic Criticality

To motivate the construction of a new auxiliary model approach in this thesis, we now write down some important outstanding questions that we will address using our approach.

1. *Can we construct an impurity-based auxiliary model approach that describes interaction-driven electronic phase transitions without imposing self-consistency?* As discussed in the preceding sections, one of the most prominent examples of auxiliary model approaches in the study of correlated fermionic models is dynamical mean-field theory (DMFT). However, solutions arising from DMFT and it's extensions such as cellular DMFT and dynamical cluster approximation suffer from a lack of transparency owing to the numerical approach towards self-consistency. To go beyond such approaches, can we explicitly construct an impurity model that maps to the local physics of some interacting lattice model, and recover the low energy physics of the latter by solving the former? Apart from being much more transparent due to the explicit nature of the construction, such an approach would also be more tractable – because of the simplicity of impurity models – than studying the full lattice model.

2. *Can such an approach map the renormalisation group flows of an auxiliary impurity model to those of a correlated lattice model?* Both impurity and lattice models, due to the presence of many-particle interactions, typically involve a hierarchy of energy scales that are coupled to one another. The framework of renormalisation group (RG) is a powerful approach to identify the "important" physics at various energy scales. In particular, it can be used to identify low-energy effective Hamiltonians that describe the essential physics in the ground state and at low temperatures, and the microscopic Hamiltonian is said to "flow" into such low-energy fixed-point Hamiltonians. Can our auxiliary model approach map the renormalisation group flows between the impurity and lattice models, at least in terms of the local physics? This would make the underlying mechanism for the RG flows as well as the various fixed-point theories much more transparent.

3. *Does such an auxiliary model approach preserve symmetries of the microscopic/effective theories of the lattice model, and obey various conservation principles such as Luttinger's theorem?* Importantly, an auxiliary model mapping for a lattice model should preserve the symmetries of the full Hamiltonian. It should also, ideally, show the emergence of conservation principles and spectral sum rules. Luttinger's theorem, as an example of such a sum rule, equates the electron density in a system with the Fermi volume in the presence of interactions, as long as the excitations are electronic. Violations in Luttinger's theorem often signal interaction-driven transition from Fermi liquids to Mott insulators or non-Fermi liquids. Can the above auxiliary model approach capture changes in Luttinger's theorem? An associated question is whether such an approach can show the emergence of Luttinger surfaces through the appearance of zeros in the one-particle Greens function. Such signatures would have topological consequences because the Luttinger sum rule is protected by topological invariants.

4. *Is it possible to describe topological transitions using such an approach?* It is becoming increasingly clear that various materials exhibit phase transitions that are not necessarily tied to the spontaneous breaking of a local symmetry and fall outside the Ginzburg-Landau-Wilson paradigm. Examples include (i) the Mott transition driven by a charge gap but without necessarily a spin gap, (ii) the Kondo breakdown QCP scenario in heavy fermion materials where the large-to-small Fermi surface transition is observed to to be separate from magnetic ordering and (iii) transition from a trivial insulator to a topological insulator driven by changes in band topology. Such transitions do not exhibit the appearance of non-zero local order parameters, and are instead often described by changes in topological invariants and entanglement patterns. One can therefore ask: is such an auxiliary model approach capable of tracking such generalised order parameters?

5. *Can such an auxiliary model approach capture the effects of multiple orbitals and momentum-space differentiation?* One of the primary difficulties with many of the numerical methods typically employed in such problems is the high computational complexity (in time as well as memory) and the inability to resolve momentum-space features to sufficient precision. The increased complexity becomes particularly important in several classes of problems such as multi-band models, which are pertinent to investigations into orbital-selective transitions, heavy fermion physics and twisted bilayer graphene, among many others. The lack of momentum-space resolution is relevant to studies of the pseudogap (PG) phase that is observed in the underdoped high-Tc superconductors; the PG is known to display partial gapping of the Fermi surface around the antinodes while remaining gapless around the nodes. The approach proposed here can benefit in this regard in two ways: the explicit construction might be helpful to avoid the computational cost of a self-consistency loop whilst providing access to a large number of momentum states, and (ii) the mapping to an impurity model makes the analysis much more tractable.

Within the thesis, we have developed a new approach towards studying correlated models with strong local interactions. The strategy goes as follows: We first identify a quantum impurity model that represents faithfully the local physics of the lattice model we are targeting. This is done by solving the impurity and studying it's phenomenology. In order to capture specific aspects of the physics, the impurity model may need to be enhanced by, say, allowing for multiple impurities, or embedding it onto a lattice instead of a continuum, etc. Having identifies the appropriate impurity model, we apply manybody translation operators to the impurity model Hamiltonian in order to restore translation invariance and map it to the target lattice Hamiltonian in the process.

For single-particle wavefunctions $\psi_i(r_i)$, the one-particle momentum operator p_i generates a one-particle translation operator $T_i(bfa)$ that translates the real-space operators associated with the particle i by the vector $\mathbf{a}$. By applying them to the dynamics of local potential wells at each site, these translation operators can be used to construct a translation-invariant Hamiltonians such as the tight-binding model; a similar application on the local wavefunctions generates delocalised translation-invariant Bloch states. Analogous to the single-particle case, one can construct a many-particle translation operator T [442] that act on an $N-$particle state $\Psi(r_1, r_2, ...r_N)$: $T(\mathbf{a}) = e^{-i\mathbf{a}\cdot\mathbf{P}} = \prod_{i=1}^{N} T_i(\mathbf{a})$, where $\mathbf{P}$ is the total momentum operator. The action of T is to shift all local operators by $\mathbf{a}$ and to multiply all $k-$space operators with a phase shift $e^{i\mathbf{a}\cdot\mathbf{k}}$.

Using such a manybody translation operator, we are able to map both the Hamiltonian as well as eigenstates between the "local" auxiliary impurity model and the translation-invariant lattice model. This leads to specific relations between various observables such as correlations and spectral functions, as well as between entanglement measures. This set of mappings can be used to reconstruct the low-energy physics of the lattice model by starting from that of the auxiliary model.

1.6.1 Important features of the "tiling" auxiliary model approach

Local transition as a diagnostic of lattice transition As mentioned in the preceding paragraphs, the approach developed within this thesis maps the low-energy physics of the lattice model to that of an impurity model with local correlation, hence rendering the problem much more tractable. This, for example, means that the presence of local Fermi liquid excitations on the impurity site marked by a broad central Kondo resonance in the local density of states corresponds to a Landau Fermi liquid phase for the lattice model. By inducing a gap in the impurity site density of states and stabilising the local moment states, one can also model a Mott insulator on the lattice model. More exotic phases such as non-Fermi liquids can be engineered by enhancing the impurity model with competing interactions that lead to a pseudogap in the impurity density of states. Taken together, this maps the renormalised groups flows of the impurity model to that of the lattice model.

Restoring translation symmetry using a many-body Blochs' theorem At the heart of the impurity-lattice model mapping are manybody translation operators T. Indeed, they serve to restore translation invariance lost by the introduction of the impurity site into the lattice: $H_{\text{lat}} = \sum_i T^\dagger(\mathbf{r}_i) H_{\text{aux}} T(\mathbf{r}_i)$. In a similar fashion, the translation operators serve to "extend" the auxiliary model eigenstates into those of the lattice model: $\Psi = \sum_i T^\dagger(\mathbf{r}_i)\psi_{\text{aux}}$. This constitutes a manybody version of Bloch's theorem [443], where local impurity model states are superposed to form delocalised wavefunctions that span the lattice. In a crude sense, this step plays the role of the self-consistent loop in numerical implementations of DMFT. This is also seen from the fact that translation operation leads to effective renormalised relations for the lattice model couplings in terms of the auxiliary model quantities. These translation operators, apart from restoring spatial translation symmetry, also introduce non-local inter-auxiliary model contributions into various correlation functions.

Mapping across Greens functions and entanglement measures One of the main results of our tiling approach is the set of mappings between real-space and momentum-space Greens functions of the auxiliary model and the lattice model. This makes it much more tractable to characterise phases on the lattice model. This mappings take into account correlations within a single impurity model as well as contributions from inter-auxiliary model connections, through the translation operators. In the absence of long-range critical fluctuations, these expressions can be truncated to terms up to a few sites away from the impurity complex, and this makes the practical computation possible. Near the transition, it becomes necessary to work with much larger number of sites.

Lattice-embedding as a route to encoding lattice geometry and band topology Another important aspect of the tiling approach is the ability to incorporate lattice effects such as momentum-space differentiation and topological features. We accomplish this by considering auxiliary model directly on the appropriate lattice. This ensures that the Hamiltonian couplings of the auxiliary model inherit the symmetries of the lattice, and the auxiliary model quantities are then sensitive to lattice effects. This, for example, has allowed us to study a momentum-space anisotropic pseudogap phase in the context of an extended Hubbard model on a 2D square lattice.

1.7 Summary Of Chapters

For the readers' convenience, we now briefly describe the content and layout of the following chapters.

The *Introduction* introduces the core themes of electronic correlation, Mott transition and auxiliary models. After discussing some qualitative features of Mottness and extracting some preliminary insights from toy models, we introduce the other major theme – auxiliary models, and describe why quantum impurity models act as auxiliary models for models with strong local correlations. We discuss the phenomenology of several classes of materials with strong local correlations, and review existing results on multiple models that describe such models, such as the Hubbard model in two and infinite dimensions, as well as lattice impurity models. We go on to describe several classes of strongly correlated materials that are likely to be described by such methods, and discuss the phenomenology of some interacting models that will be investigated within the thesis. Near the end, we describe some of the features of the new auxiliary model developed within the thesis and list some of the questions addressed by it. We end by introducing various features of entanglement and its role in the holographic principle, and list some questions addressed by our work on entanglement renormalisation work in a simple model of metal-insulator transition driven by a mass gap.

After the introduction, the *Methods and Preliminaries* chapter describes the various techniques and some preliminary ideas employed throughout the thesis. The chapter begins by deriving the unitary renormalisation group formalism and describes it's various properties and a practical protocol for implementing it. That is followed by demonstrations of the unitary RG on three simple models - a star graph model of spins in a magnetic field, the two-channel Kondo model, and the problem of a particle in a periodic potential. Through these problems, various aspects of the method are clarified. In order to provide more insights, the chapter also details how the unitary RG method is related to other RG-based methods that rely on canonical transformations, such as Poor Man's scaling, Schrieffer-Wolff transformation and CUT (continuous unitary transformation) RG. Apart from the URG, the chapter also describes an iterative diagonalisation method that has been used within this thesis to compute various correlation functions and entanglement measures from the hierarchy of Hamiltonians obtained from the URG. Since various entanglement measures have been used throughout the later chapters, we have also added a section on various entanglement measures here - what they mean, what are their inter-relations, why they are important and how one can compute some of them. The chapter concludes by describing some of the other strategies that have been employed to reduce computational costs in various projects.

In Chapter 3, *Kondo Frustration via Charge Fluctuations: A Route to Mott Localisation in Infinite Dimensions*, we study our first impurity model, the extended Anderson model with local correlation U_b in the conduction bath. We demonstrate that this model hosts a local version of the

Mott metal-insulator transition on the Bethe lattice in infinite dimensions, as seen from DMFT. At a critical value of the ratio of U_b and the impurity-bath hybridisation, the effective impurity model shows a transition from a Kondo screened phase into an unscreened local moment phase. For the case of attractive local bath correlations, the model sheds new light on several aspects of the DMFT phase diagram. For example, the T = 0 metal-to-insulator quantum phase transition (QPT) is preceded by an excited state QPT (ESQPT) where the local moment eigenstates are emergent in the low-lying spectrum. Long-ranged fluctuations are observed near both the QPT and ESQPT, suggesting that they are the origin of the quantum critical scaling observed recently at high temperatures in DMFT simulations. The $T = 0$ gapless excitations at the quantum critical point display particle-hole interconversion processes, and exhibit power-law behaviour in self-energies and two-particle correlations. These are signatures of non-Fermi liquid behaviour that emerge from the partial breakdown of the Kondo screening. A many-body perturbation theoretic treatment of the Hubbard sidebands reveals that they are comprised of holon-doublon excitations created by the hybridisation of the impurity site with the conduction bath. These excitations consist of decoupled local Fermi liquids for the holons and doublons at the lowest order, which, at higher orders, become coupled via correlated holon-doublon scattering between impurity and bath.

In Chapter 4, *A New Auxiliary Model Approach for Interacting Electrons: Mapping Impurity Models To Lattice Models*, we lay out the details of the auxiliary model method developed to map impurity models to lattice models. We first motivate the general idea behind auxiliary model methods. This is followed by a construction of the manybody translation operators. We use this, along with a manybody Bloch's theorem, to construct lattice models from quantum impurity models and relate their eigenstates. This then allows us to map correlations functions, self-energies and entanglement measures across models. In particular, we demonstrate how the presence of the conduction bath introduces $k-$space resolution in various quantities.

In Chapter 5, *Mott Criticality as the Confinement Transition of of a Pseudogap-Mott Metal: Insights on Mott Transition and Pseudogap in Two Dimensions*, we apply the auxiliary model tiling method to a lattice-embedded extended SIAM. Applying a many-body tiling scheme to the fixed-point impurity model uncovers a lattice model with electron interactions and Kondo physics. At half-filling, the interplay between Kondo screening and bath charge fluctuations in the impurity model leads to Fermi liquid breakdown. This reveals a pseudogap phase characterized by a non-Fermi liquid (the Mott metal) residing on nodal arcs, gapped antinodal regions of the Fermi surface (Luttinger surfaces), and an anomalous scaling of the electronic scattering rate with frequency. The eventual confinement of holon-doublon excitations of this exotic metal obtains a continuous transition into the Mott insulator. Our results identify the pseudogap as a distinct long-range entangled quantum phase, and offer a new route to Mott criticality beyond the paradigm of local quantum criticality. The phase appears to encode multipartite entanglement, as seen from a large value of the quantum Fisher information. The critical point at the pseudogap-Mott insulator boundary is found to be described by the exactly-solvable Hatsugai-Kohmoto model. The emergence of Luttinger surfaces alters the anomaly structure associated with the generalized symmetry of the FS [44–46]. The corresponding topological charge is defined via the Luttinger–Ward functional. This charge is modified by Green's function zeros, signalling a shift in the underlying anomaly. Within our framework, this reorganization grants topological protection to the RG flows terminating in the PG regime, ensuring the stability of its NFL excitations.

In Chapter 8, *Competing tendencies In Heavy Fermion Systems: Kondo screening vs. Mott localisation*, we study a periodic Anderson model where we treat both the $c-$ and $f-$layers as extended Hubbard models, and analyse it using the tiling approach developed and implemented in the earlier chapters. We first study the case of half-filling and obtain a Kondo insulator at large values of the inter-band hybridisation. We also study the more interesting case of a Mott metal in the $f-$layer hybridising with the $c-$electrons.

In Chapter 7, *Holography of Entanglement in 2D Free Fermions: Emergent Geometry and Fermionic Criticality*, we provide an explicit demonstration of the holographic principle in the form of a simple and tractable exact holographic mapping (EHM) for a free fermion system with and without a mass gap. Our program yields analytic relations between the RG flow parameters and their dual bulk quantities such as distances and curvature. This also allows us to study the consequences of a critical Fermi surface (lying precisely at the quantum phase transition) on the nature of the holographic space.

We conclude in the final chapter by summarising the main results obtained within the thesis, in terms of the methods developed as well as insights obtained into the behaviour of various models. We describe the impact of these results on the field, the broad conclusions emerging from them and some open questions resulting from the work.

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